



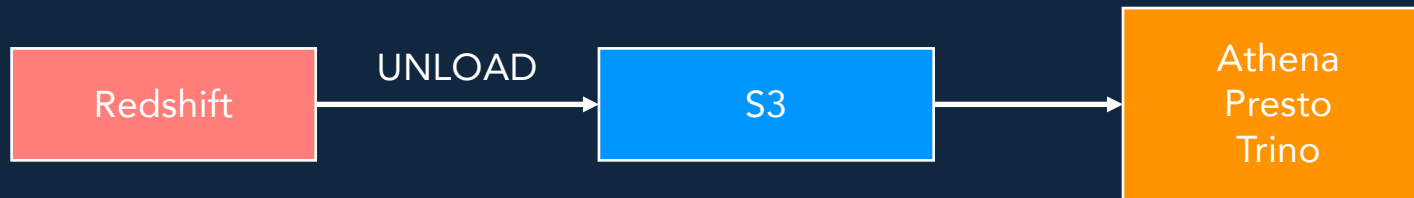
Feature Usages

COPY (Load Data in Batch)



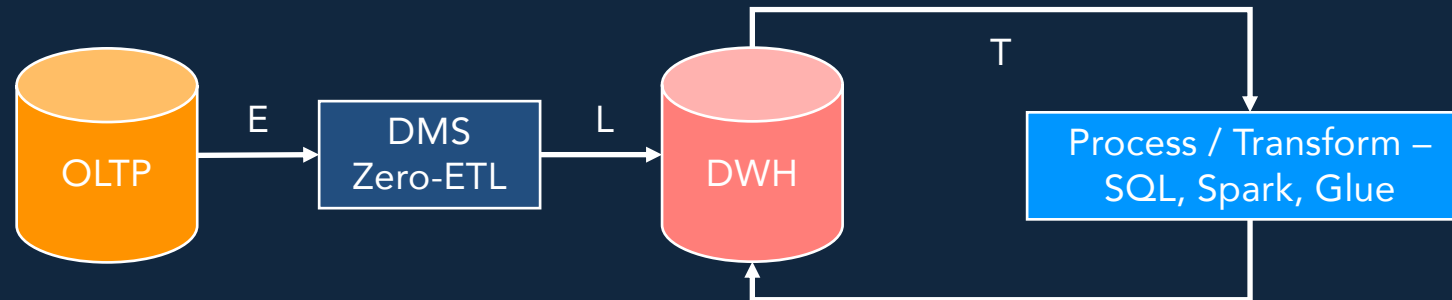
- Load data without any transformation.
- Batch load (using one or multiple files).
- Transform data and store in S3 before loading to Redshift.
- CSV, JSON, Parquet, Orc, Avro.
- Continuous file ingestion from S3 (preview).

UNLOAD (Extract & Archive)



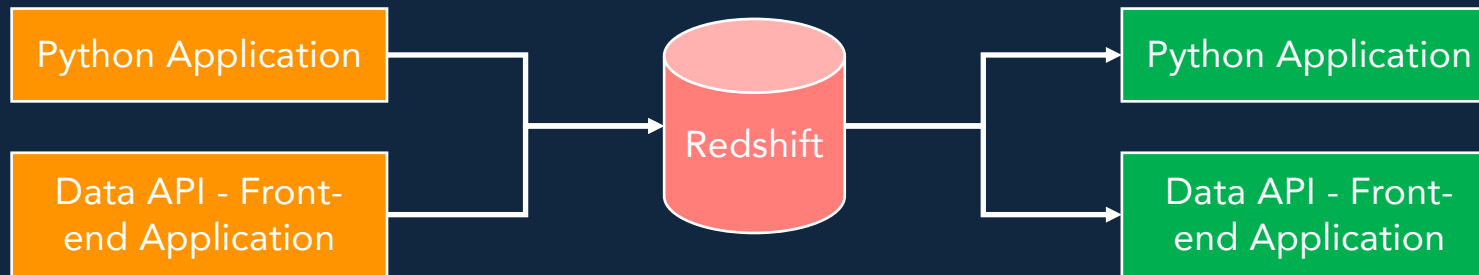
- Extract data from a Redshift table to S3.
- CSV, JSON and Parquet.
- Archive – Extract records to S3 → Delete from table.
- Use other services to read archived data, if needed.

DMS and Zero-ETL (ELT)



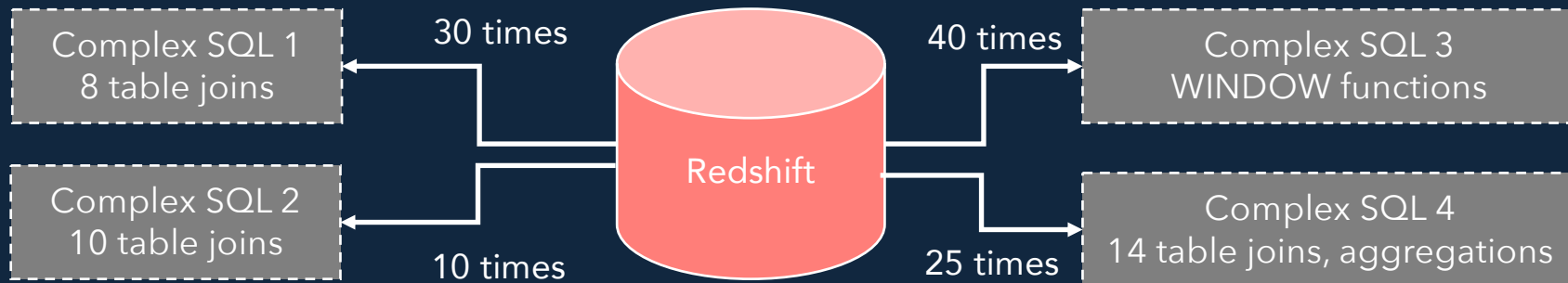
- Extract and Load from OLTP sources to DWH tables.
- Transform or process data after loading.
- Load processed data back to other Redshift tables.

Python, Data API (Application & DWH)



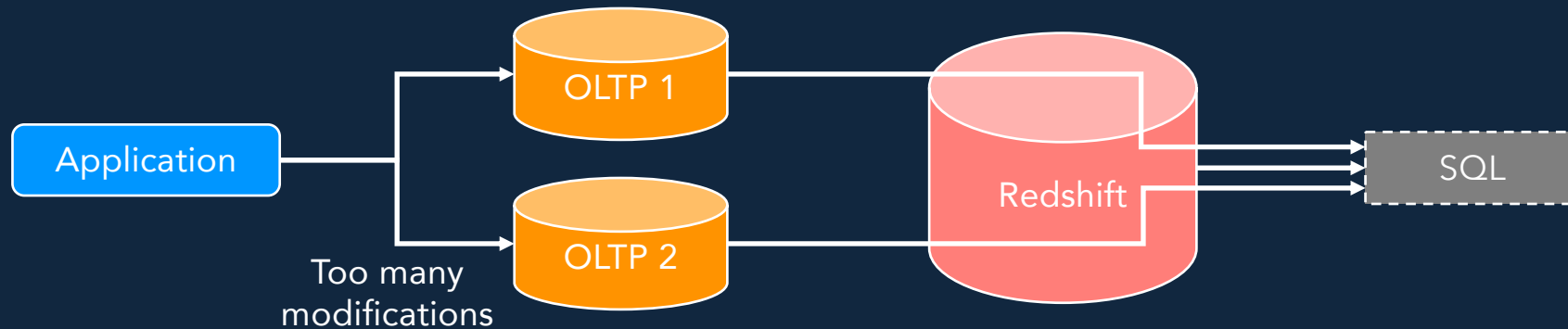
- Write or Read data from application without any BI tools.
- Writes do not happen that often.

Materialized Views (Materialize Result)



- Every time CPUs are used to execute the complex queries.
- Store the results in disk.
- Read directly from disk without complex computation.

Federation (Bring OLTP and OLAP in One Place)



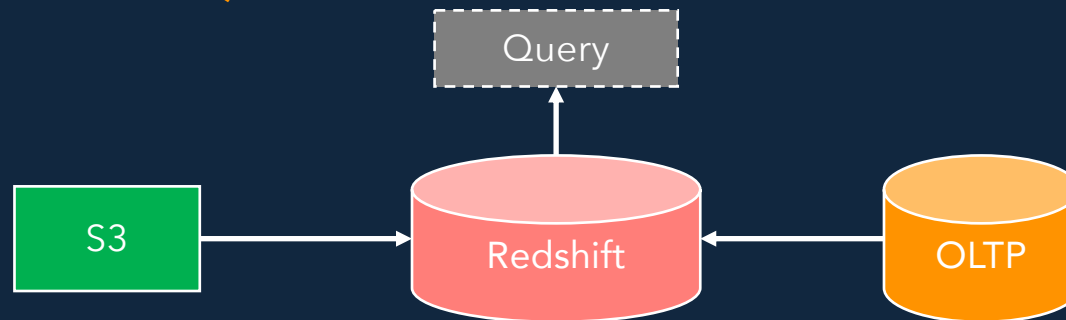
- Use a single pane of glass to query OLTP and DWH.
- Capture near real-time OLTP modifications in queries on Redshift.
- No need to move live modifications to Redshift. Do it later using batches.

Streaming in DWH ??



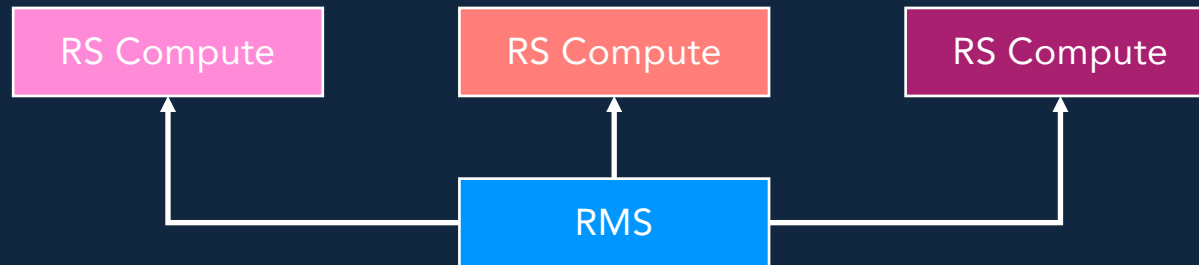
- Continuous ingestion with near real-time query capabilities.
- Query and BI tools need to monitor continuous dashboards.
- Append table and MV on top of it might be important.
- How to handle schema evolution?

Spectrum (Bring Data Lake & DWH in One Place)



- Query Data Lake without moving it to Redshift.
- Query archived data (stored using UNLOAD).
- Lot of incoming files at frequent interval.
- Query OLTP, DWH and Data Lake from Redshift.

Data Sharing (Don't Just Move Data)



- Use the same cluster for multiple teams and use cases.
- No need to move data across clusters.
- Can implement cohesive data mart.

